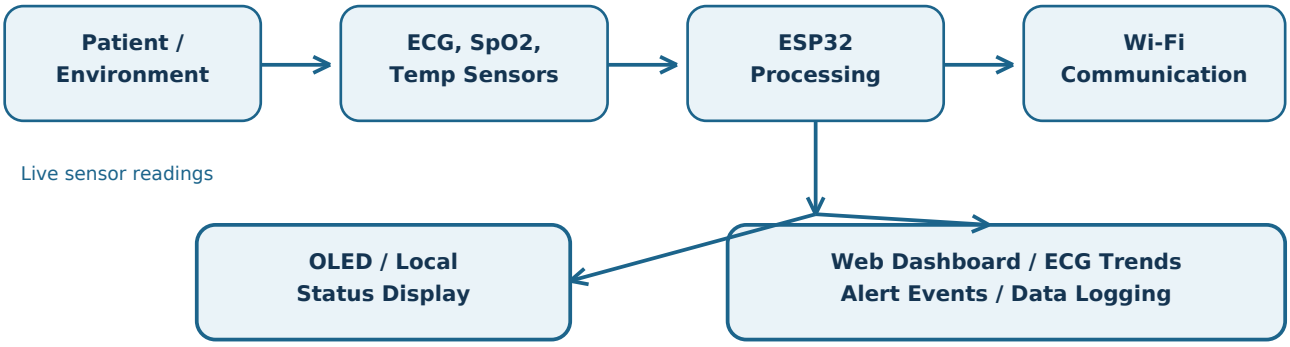
# SMARTCARE

ESP32-Based IoT Patient Monitoring & Intelligent Emergency Alert System

## METHODOLOGY

SMARTCARE is a proposed portable, ESP32-based IoT monitoring prototype. An AD8232 module will acquire ECG signals, while a MAX30102 sensor will estimate pulse rate and blood oxygen saturation (SpO2). An NTC sensor will support temperature measurement, and a BME280 will capture environmental temperature and humidity. The ESP32 will collect sensor readings, apply basic filtering and signal-quality checks, and communicate through Wi-Fi. Live values and ECG waveforms will appear on an OLED display and a browser-based dashboard. Historical measurements and alert events will be logged for trend analysis. Configurable thresholds and persistence checks will distinguish sustained abnormal readings from isolated noisy samples before activating dashboard warnings and a buzzer. The prototype will be tested for sensor response, dashboard updates, alert behavior and operational stability.

## PROPOSED BLOCK DIAGRAM




## PROPOSED OUTCOME

The proposed outcome is a compact, battery-powered educational prototype demonstrating integrated monitoring of ECG, heart rate, SpO2 and temperature. It is expected to provide real-time visualization, historical trends, signal-quality indications and configurable warning/critical alerts through a local display and web dashboard. The system emphasizes affordability, portability and modular expansion toward future cloud or mobile integration. It is intended for engineering demonstration only, not medical diagnosis or clinical decision-making.